

Exp.No-5: MIMO System Simulation with BER Analysis

CODE IMPLEMENTATION:

```
MIMO_System_Simulation_BER_Analysis.m × +
/MATLAB Drive/MIMO_System_Simulation_BER_Analysis.m
1  clc;
2  clear;
3  close all;
4
5  snr = 0:2:20;
6  BER = 0.5*erfc(sqrt(10.^(snr/10)));
7
8  capacity = log2(1 + 10.^(snr/10));
9
10 selected_snr = 10;
11 ber_value = 0.5*erfc(sqrt(10^(selected_snr/10)));
12 capacity_value = log2(1 + 10^(selected_snr/10));
13
14 fprintf('===== MIMO SYSTEM SIMULATION =====\n\n');
15 fprintf('Selected SNR = %d dB\n',selected_snr);
16 fprintf('Bit Error Rate (BER) = %.6f\n',ber_value);
17 fprintf('Channel Capacity = %.2f bps/Hz\n',capacity_value);
18
19 figure;
20
21 subplot(2,2,1);
22 semilogy(snr,BER,'b-o','LineWidth',2);
23 grid on;
24 title('Objective 1');
25 xlabel('SNR (dB)');
26 ylabel('BER');
27
```

OUTPUT:

